

King Fahad Central Hospital

Early Mobilization Protocol for Acute Stroke Patients

Aligned with AHA/ASA Guidelines for Early Management of Acute Ischemic Stroke, the AHA/ASA scientific statement on early mobilization, and AVERT trial evidence

1. Purpose & Guiding Principle

Mobilize stroke patients as soon as they are medically stable, using a structured, incremental, and individualized approach. Evidence from the AVERT trial showed that very early, high-frequency, high-dose mobilization within 24 hours of onset was associated with worse functional outcomes at 3 months compared with usual care. Current guidance favors mobilization out of bed within 24–48 hours for most patients, avoiding very early high-intensity protocols, while still avoiding prolonged bed rest.

- ☐ **Do not delay mobilization due to routine convenience — but do not pursue frequent, high-dose, very-early (<24h) out-of-bed mobilization outside of a structured/individualized protocol**
- ☐ Individualize timing and intensity based on stroke severity, stroke subtype, hemodynamic stability, and any acute intervention performed (thrombolysis, thrombectomy, hemicraniectomy)

2. Pre-Mobilization Safety Screening (complete before each session)

- ☐ Patient is medically stable: no new or worsening neurologic deficit in preceding 2 hours
- ☐ Systolic BP within physician-ordered target range (see admission/post-procedure orders); no symptomatic hypotension
- ☐ Heart rate 40–120 bpm and no new dysrhythmia
- ☐ SpO₂ ≥ 92% on current oxygen support
- ☐ Temperature < 38.5°C
- ☐ Blood glucose within acceptable range (not severely hypo- or hyperglycemic)
- ☐ No unstable cardiac or respiratory condition (e.g., active ischemia, uncontrolled arrhythmia, respiratory distress)
- ☐ Pain adequately controlled
- ☐ Patient alert enough to participate and follow simple commands (or mobilization plan adjusted for level of consciousness)

If any item fails, defer mobilization, document reason, and reassess at next scheduled interval or per nursing/PT judgment.

3. Condition-Specific Timing Considerations

Clinical scenario	Mobilization timing guidance
Uncomplicated ischemic stroke, hemodynamically stable	Begin low-dose, out-of-bed mobilization within 24–48h once safety screen passed; avoid frequent high-intensity sessions in first 24h
Post IV thrombolysis (alteplase/tenecteplase)	Follow post-tPA neuro/vital sign monitoring schedule; mobilization out of bed typically deferred until after the high-frequency monitoring period and repeat imaging clears hemorrhage, per protocol
Post mechanical thrombectomy	Bed rest/limb immobilization per vascular access closure method first (see access-site protocol); once access site cleared and post-procedure imaging excludes hemorrhage, proceed with incremental mobilization
Intracerebral hemorrhage	More conservative, individualized approach; mobilize once hematoma stability confirmed on imaging and BP/ICP considerations addressed — per treating neurologist/neurosurgeon
Large-territory infarct / malignant edema risk	Individualize; close neuro monitoring during mobilization; involve neurosurgery if decompressive craniectomy being considered
Post decompressive hemicraniectomy	Per neurosurgical direction; skull protection (helmet) considerations during out-of-bed activity
ICU / mechanically ventilated patients	Structured ICU early-mobility protocol per critical care team; coordinate with sedation/ventilator weaning plan

4. Tiered Mobility Protocol (PT/OT/Nursing-Directed)

Progress sequentially as tolerated; do not skip tiers for patients with significant deficits without PT/OT assessment. Reassess safety screen before each tier advancement.

Tier	Activity
Tier 1 — Bed level	Active/passive range-of-motion exercises in bed; positioning changes q2h; head-of-bed elevation as ordered; sitting supported in bed
Tier 2 — Sitting	Sit on edge of bed with supervision/assistance; sitting balance training; transfer to bedside chair
Tier 3 — Standing/Transfer	Sit-to-stand practice with assistance/gait belt; standing balance; pivot transfers
Tier 4 — Ambulation	Assisted ambulation with appropriate device (gait belt, walker, cane) and staffing per fall-risk assessment
Tier 5 — Independent mobility	Progress toward independent ambulation and functional tasks per PT/OT goals; prepare for discharge mobility level

5. Monitoring During Mobilization

- ☐ Check vital signs (BP, HR, SpO2) immediately before and after each mobilization session; check symptoms (dizziness, chest pain, dyspnea) during activity
- ☐ Use gait belt and appropriate number of staff per fall-risk assessment
- ☐ Have suction and emergency equipment accessible per unit policy for high-risk patients
- ☐ Document tier achieved, distance/duration, assistance level, and any adverse events

5a. Stop / Do-Not-Advance Criteria — halt session and notify physician if any occur

- ☐ Drop in SBP > 20 mmHg from resting baseline with symptoms (dizziness, near-syncope)
- ☐ New or worsening focal neurologic deficit
- ☐ SpO2 falls below 90% despite supplemental oxygen
- ☐ Heart rate <40 or >130 bpm, or new symptomatic arrhythmia
- ☐ Patient reports chest pain, severe dyspnea, or severe headache
- ☐ Signs of access-site bleeding (post-thrombectomy) or hematoma expansion
- ☐ Patient distress, refusal, or inability to safely participate

6. Roles & Responsibilities

- ☐ Nursing: bed-level mobility (Tier 1), positioning q2h, safety screen prior to any out-of-bed activity, vital sign monitoring
- ☐ Physical Therapy: formal mobility/gait assessment within 24–48h of admission; establishes and progresses tiered mobility plan
- ☐ Occupational Therapy: functional/transfer training, activities of daily living, cognitive-perceptual screening affecting safe mobility
- ☐ Physician/Advanced Practice Provider: sets BP/hemodynamic parameters, clears mobilization after procedures, addresses any stop-criteria events
- ☐ Speech-Language Pathology: coordinates with mobility plan if aspiration/dysphagia precautions affect positioning

7. Documentation & Quality Metrics

- ☐ Time from admission (or procedure) to first out-of-bed mobilization documented
- ☐ PT evaluation completed within 24–48h of admission (core measure)
- ☐ Highest mobility tier achieved documented each shift/day
- ☐ Any mobilization-related adverse events (falls, symptomatic hypotension, desaturation) logged and reviewed
- ☐ Mobility milestones tracked toward discharge planning and rehabilitation referral

Physician / Team Authorization

Physician Signature: _____ Date/Time: _____

BP parameters for mobilization: SBP < _____ / DBP < _____ Other restrictions: _____

References: AHA/ASA scientific statement on early mobilization after acute stroke; 2019 AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; AVERT trial (Bernhardt et al., Lancet 2015); AHA/ASA Get With The Guidelines–Stroke measures (PT evaluation, rehabilitation assessment); The Joint Commission Primary/Comprehensive Stroke Center certification requirements. This template should be re-verified against the current guideline version at the time of institutional adoption.